

Session 4: Data Retrieval with SELECTTrainer Coverage:SELECT queries, filtering, sorting, aliases.Aggregate functions (SUM, AVG, COUNT).Demo: Generate “total expenses” by category.Pending

Tasks

- 1) Create a table called Playlist with columns: id, song_name, artist, genre, and play_count. Insert at least 5 sample records representing your favorite songs from Spotify.

Query:

Run SQL query/queries on database my database: 

```
1 CREATE TABLE Playlist (  
2     id INT AUTO_INCREMENT PRIMARY KEY,  
3     song_name VARCHAR(100),  
4     artist VARCHAR(100),  
5     genre VARCHAR(50),  
6     play_count INT  
7 );  
8  
9 INSERT INTO Playlist (song_name, artist, genre, play_count)  
10 VALUES  
11 ('Blinding Lights', 'The Weeknd', 'Pop', 150),  
12 ('Shape of You', 'Ed Sheeran', 'Pop', 200),  
13 ('Perfect', 'Ed Sheeran', 'Romantic', 175),  
14 ('Believer', 'Imagine Dragons', 'Rock', 120),  
15 ('Kesariya', 'Arijit Singh', 'Bollywood', 180);|
```

Output:

☐ Show all | Number of rows: 25 ▾ | Filter rows: | Sort by key

Extra options

				▼ Id	song_name	artist	genre	play_count
☐	 Edit	 Copy	 Delete	1	Blinding Lights	The Weeknd	Pop	150
☐	 Edit	 Copy	 Delete	2	Shape of You	Ed Sheeran	Pop	200
☐	 Edit	 Copy	 Delete	3	Perfect	Ed Sheeran	Romantic	175
☐	 Edit	 Copy	 Delete	4	Believer	Imagine Dragons	Rock	120
☐	 Edit	 Copy	 Delete	5	Kesariya	Arijit Singh	Bollywood	180

2) Write a SELECT query to display only the song_name and artist from the Playlist table, but show the artist column as 'Singer' using an alias.

Query:

```
SELECT song_name, artist AS Singer
FROM playlists;
```

Output:

						☐ Profiling [
		☐ Show all	Number of rows:	25 ▾	Filter rows:	Search this table
Extra options						
				▼ song_name	Singer	
☐	Edit	Copy	Delete	Blinding Lights	The Weeknd	
☐	Edit	Copy	Delete	Shape of You	Ed Sheeran	
☐	Edit	Copy	Delete	Perfect	Ed Sheeran	
☐	Edit	Copy	Delete	Believer	Imagine Dragons	
☐	Edit	Copy	Delete	Kesariya	Arijit Singh	

3) Write a SELECT query to find all songs in the Playlist table where the genre is 'Pop' and play_count is greater than 100. Sort the results by play_count in descending order.

Query:

Run SQL query/queries on table `my database.playli`:

```
1 SELECT*
2 FROM playlists
3 WHERE genre='pop'
4 AND play_count>100
5 ORDER BY play_count DESC;
```

Output:

☐ Show all | Number of rows: 25 ▾ | Filter rows: | Sort by key:

Extra options

				id	song_name	artist	genre	play_count	1
☐	 Edit	 Copy	 Delete	2	Shape of You	Ed Sheeran	Pop	200	
☐	 Edit	 Copy	 Delete	1	Blinding Lights	The Weeknd	Pop	150	

- 4) Use the COUNT aggregate function to find out how many songs in your Playlist table belong to the genre 'pop'.

Query:

Run SQL query/queries on table my_database.playlists

```
1 SELECT COUNT(*) AS total_pop_songs
2 FROM playlists
3 WHERE genre='pop';|
```

Output:



The screenshot shows a SQL query result in a web interface. The query is: `SELECT COUNT(*) AS total_pop_songs FROM playlists WHERE genre='pop';`. The result is displayed in a table with one column named `total_pop_songs` and one row with the value `2`. There is an "Extra options" button above the table.

total_pop_songs
2

5) Write a SELECT query that shows the total number of plays (SUM of play_count) for each genre in your Playlist table, grouped by genre.

Hint: Use GROUP BY with the SUM function to aggregate play counts by genre.

Query:

Run SQL query/queries on table my database.playlists: ?

```
1 SELECT genre, SUM(play_count) AS total_plays
2 FROM playlists
3 GROUP BY genre;
```

Output:

☐ Show all | Number of rows: 25 v | Filter rows: Search

Extra options

	genre	total_plays
 Edit  Copy  Delete	Bollywood	180
 Edit  Copy  Delete	Pop	350
 Edit  Copy  Delete	Rock	120
 Edit  Copy  Delete	Romantic	175